How the score works

Exposio, MSc prototype, Obuda University. Every number below comes from a live run of the scoring code.

Why there are two clocks

My first version discounted every breach by its age. It scored a friend's address 3 out of 100, even though four breaches had already made his date of birth, name, nationality, phone number and home address public. His breaches were four to eight years old, so the decay term had eaten almost all of it.

The mistake was applying one rule to two different things. A password can be changed, and once it is, the stolen copy is worthless, so that risk really does fade. A date of birth cannot be changed. It is as accurate today as the day it leaked.

So the model keeps two separate accounts. The **fast clock** covers passwords and payment data: summed per breach, weight halving about every 347 days. The **slow clock** covers names, dates of birth, addresses and the like: combined as a union and never discounted, because one fact leaked twice is still one fact.

$$B_{raw} = \Sigma\ sev_i \times e^{-\lambda \cdot age} + W \times [\ 1 - \Pi\ (1 - w_k)\ ]$$
$$score = \min(1,\ B_{raw} / B_{MAX}) \times 100$$

Constant	Value	
λ	0.002 / day	fast-clock decay, half-life 347 days
W	0.70	weight of the permanent half
B_{MAX}	0.80	top of the scale
undated age	730 days	used when no date can be found

These constants are chosen, not fitted. No public dataset records which breach victims were later defrauded, so there is nothing to calibrate against. The last section shows what happens if they are wrong.

A worked example

Four breaches found for one address.

Breach	Tier	Severity	Age (days)	Recency	Fast clock	Permanent attributes
Ticketmaster	financial	1.00	891	0.1683	0.1683	name
Combolist 2023	credentials	0.70	730 (assumed)	0.2322	0.1626	—
Canva	email	0.20	2714	0.0044	0.0000	name, username, location
LinkedIn	credentials	0.70	3815	0.0005	0.0003	phone

Only breaches that leaked credentials or payment data count on the fast clock: 0.1683 + 0.1626 + 0.0000 + 0.0003 = **0.3312**.

The union of permanent attributes across all four is name, phone, username and location. Each counts once:

$1 - (1-0.08)(1-0.14)(1-0.07)(1-0.05) = 0.3010$, then $0.70 \times 0.3010 =$ **0.2107**

	Raw	Points	Share
Fast clock, fades as passwords are changed	0.3312	41.4	61%
Permanent, no action removes it	0.2107	26.3	39%
Total, divided by 0.80 and scaled to 100	0.5419	68	band: high

The score is **68 out of 100**, and **26 of those points cannot be removed by any action**. The interface shows that figure as the floor the score settles on if the user changes nothing.

When a breach has no date

Some sources give no date at all, others only a month. The model tries three times to find one before assuming anything. It uses an exact date if given, the first of the month if only a month is given, and otherwise looks the breach up in a bundled catalogue of 1,048 known breaches. Only if all three fail does it assume 730 days, and it marks the record as assumed.

The assumption only touches the fast clock:

Case	Score	Fast clock	Permanent
Dated, credentials only	31	31.3	0.0
Undated, credentials only	20	20.3	0.0
Dated, identity only	30	0.0	29.5
Undated, identity only	30	0.0	29.5

Losing the date costs a credential breach about a third of its weight and costs a permanent one nothing. Under the old single-clock model the whole score rested on this guess. Splitting the clocks raised the rank correlation across the assumption sweep from 0.769 to 0.951.

If the constants are wrong

I perturbed all fourteen weights at once across 2,000 synthetic profiles:

Perturbation	Mean rank correlation	Minimum	Band agreement
±10%	0.999	0.998	97.8%
±20%	0.998	0.994	95.0%
±30%	0.995	0.985	91.8%

I also compared the model against seven simpler ones. The closest, which uses only each person's worst breach, agreed with mine on 83.8% of band assignments before the revision and 49.8% after it. That drop is the evidence that adding breaches up does something a single breach cannot.

The score ranks a person's own exposures against each other and orders what to fix first. It is not a probability of harm, and I do not present it as one.